

NFL Fantasy Football High-Scoring Game Prediction

CS 4700 Introduction to Machine Learning Final Project

Student: Tuna Boncu

Instructor: Prof. Xin Wang

1. Project Topic and Research Question

This project uses machine learning to predict whether an NFL offensive player will have a high-scoring fantasy football game. The research question is: Can a model predict whether a player will score at least 10 PPR fantasy points in a weekly game using position, team, experience, prior averages, and recent trend statistics?

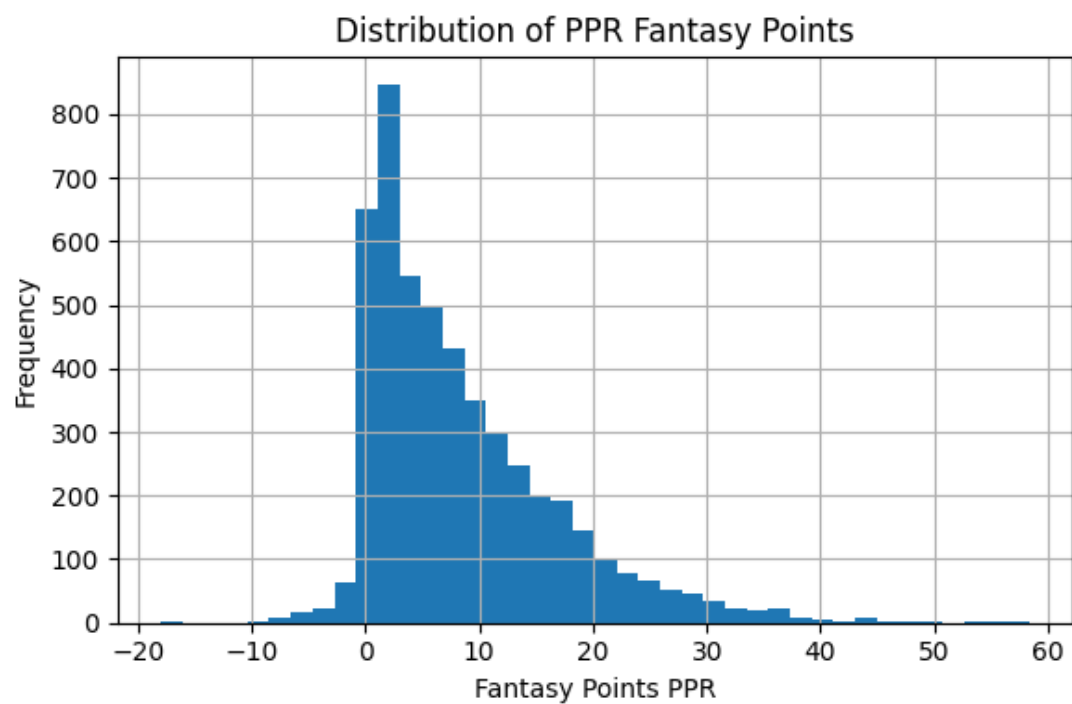
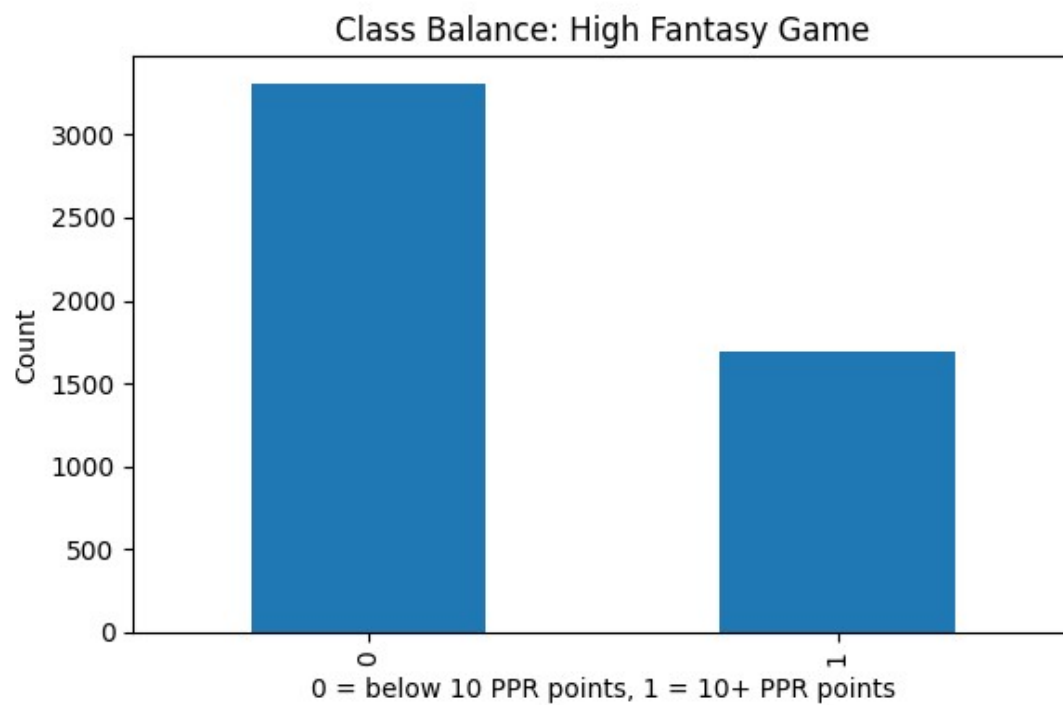
2. Dataset

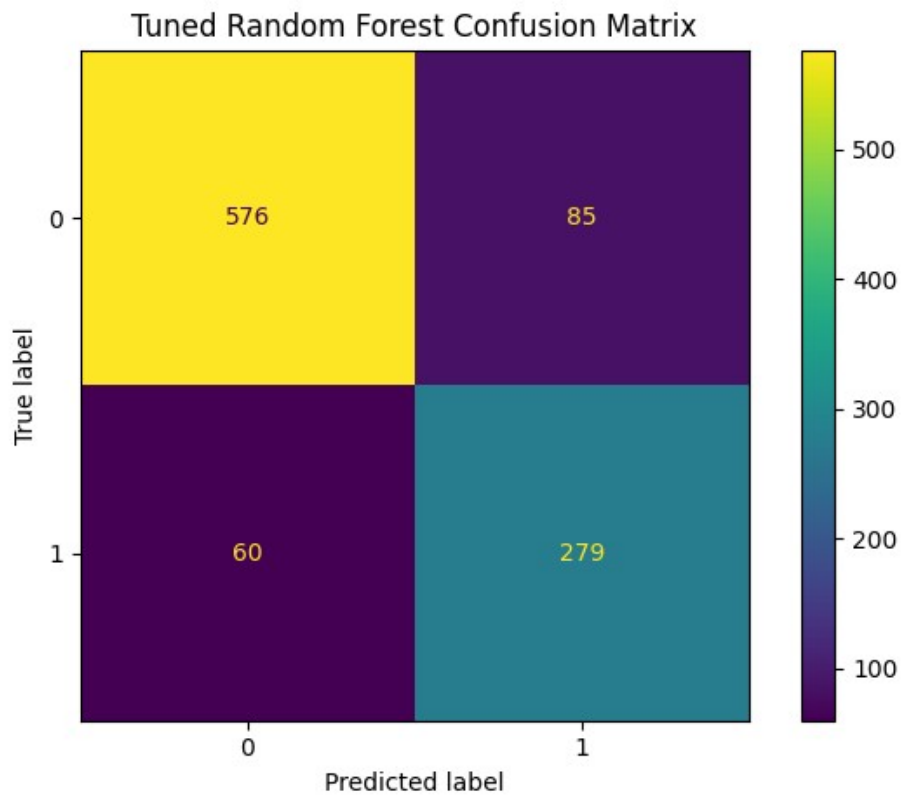
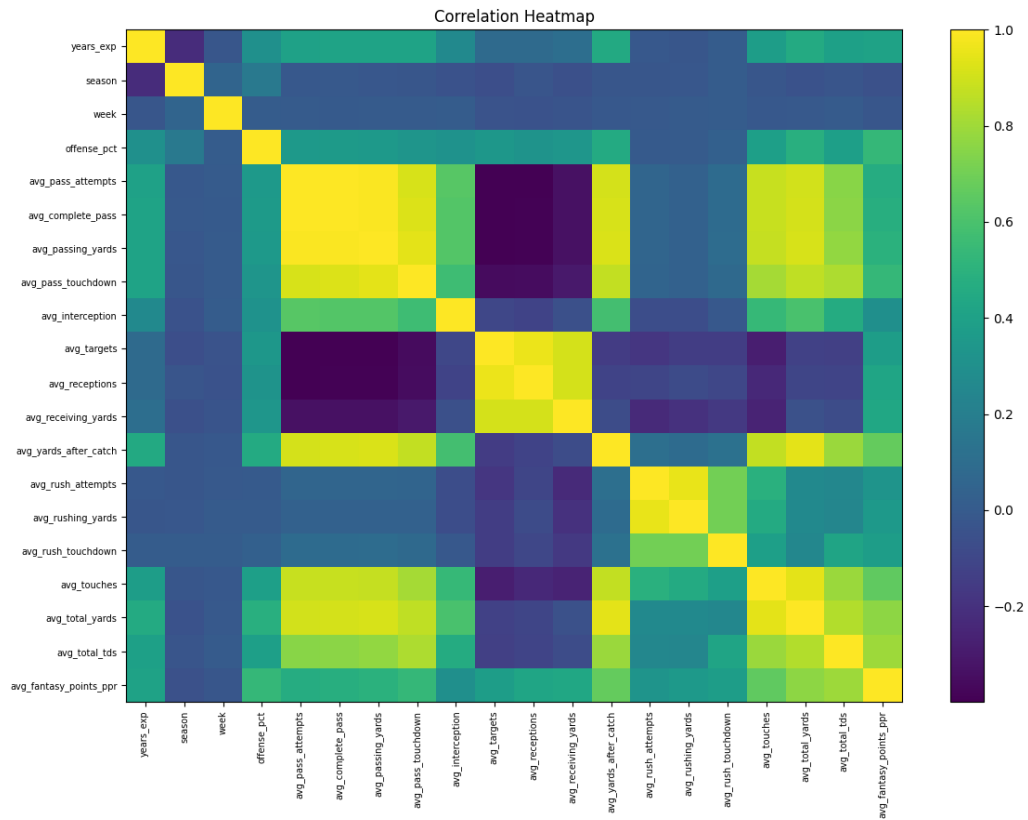
The dataset is an NFL offensive statistics archive containing weekly and yearly player and team statistics. The main file used in this project is `weekly_player_stats_offense.csv`. The project focuses on weekly offensive player records because the goal is to predict a weekly fantasy outcome.

The target variable was created from `fantasy_points_ppr`. A player was labeled as 1 if he scored at least 10 PPR fantasy points, and 0 otherwise. This converted the problem into a binary classification task.

3. Data Inspection and EDA

The data was loaded with pandas and inspected for shape, column names, data types, missing values, and target distribution. The project created visualizations for class balance, the distribution of PPR fantasy points, correlations among numeric features, and the final confusion matrix.





4. Data Preparation

Missing numerical values were handled with median imputation. Missing categorical values were handled with most-frequent imputation. Categorical features such as team, position, and position group were encoded with one-hot encoding. Numerical features were scaled using StandardScaler. These steps were placed inside scikit-learn pipelines to avoid data leakage.

5. Feature Engineering

The main engineered feature was the binary target variable `high_fantasy_game`. The model also used pre-existing average and trend variables from the dataset, such as `avg_fantasy_points_ppr`, `avg_touches`, `avg_total_yards`, `prev_4_trend_fantasy_points_ppr`, `prev_4_trend_touches`, and `prev_4_trend_total_yards`. Direct current-game fantasy points were removed from the feature set to prevent target leakage.

6. Modeling Approach

The baseline model was Logistic Regression because it is simple, interpretable, and works well as a starting classifier. The final model was a Random Forest Classifier because it can capture nonlinear relationships and handle mixed feature types after preprocessing. Hyperparameters were tuned using GridSearchCV with 3-fold cross-validation.

7. Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.845	0.737	0.844	0.787	0.912
Tuned Random Forest	0.855	0.766	0.823	0.794	0.926

The tuned Random Forest model performed best overall, with 0.855 accuracy, 0.794 F1 score, and 0.926 ROC-AUC. Compared to Logistic Regression, it improved accuracy, precision, F1 score, and ROC-AUC.

8. Error Analysis

The confusion matrix showed that the final model correctly identified many high-scoring and low-scoring performances. Most errors likely came from unpredictable NFL factors that were not fully included in the data, such as injuries, game script, weather, defensive matchups, and sudden changes in player usage.

9. Strengths and Limitations

Strengths: The project uses a real sports dataset, builds a full preprocessing pipeline, compares a baseline and final model, uses cross-validation, and includes visual error analysis. The Random Forest model handled nonlinear patterns better than Logistic Regression.

Limitations: The model used a sample of the dataset for faster training. It does not include all real-world factors such as injuries, weather, betting lines, or opponent defensive strength. Fantasy football is also naturally noisy because player usage can change suddenly.

10. Conclusion

This project showed that machine learning can predict high-scoring NFL fantasy performances with useful accuracy. The final Random Forest model outperformed the baseline model and showed that prior averages and recent trends are valuable features. Future improvements could include using the full dataset, adding injury reports, adding opponent defense statistics, and testing more advanced models such as XGBoost or neural networks.

References

CS 4700 Final Project Handout.

CS 4700 lecture notes on machine learning history, VC dimension, PAC learning, overfitting, regularization, and neural network training.

NFL offensive statistics dataset provided for the project.